

SAUHARDHA KAFLE

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SKILLS

Languages: Rust, TypeScript, Python
Backend: Actix-web, NodeJS, ExpressJS, AIOHTTP, Flask, FastAPI
Frontend: NextJS, ReactJS, TailwindCSS,
Databases & Tools: Redis, MySQL, PostgreSQL, MongoDB, Appwrite
AI/ML: Deep Learning, Natural Language Processing, RAG
Infra: Docker, gRPC, IPC, webRTC, websocket

EXPERIENCE

AI Fellowship – KonnectCraft

2026

Developed an ingestion pipeline for an graphical RAG application that extracts intelligence from the unstructured dataset. Also built an embeddable chatbot system enabling clients to integrate their custom RAG-powered assistants into their applications

Events

- 2025 **Technical Lead – ITMeet**
Led the technical team in developing AR-based marketing, an AR venue path finder, and the event website, while ensuring reliable internet infrastructure throughout the hackathon and venue.
- 2024 Event Manager – IdeaPitching | Technical Member – ITMeet

Volunteer

- 2023 KURC Volunteer | KUCC Volunteer

EDUCATION

Kathmandu University

Dec 2022 – Dec 2026

Bachelor in Computer Engineering

PROJECTS

[nutshell](#) | TypeScript, Rust, Python, NextJS

- A developer platform to build and deploy web application on cloud. A platform where we can host our website in few clicks similar to vercel.
- Uses **Cloudflare's R2 storage**, **docker** to build the code, **redis** for information flow, **Rust** to handle the request, **NextJS** for frontend and **prisma postgres** for database. Completes overall hosting process within (~60 seconds.)

[CodeRoom](#) | NodeJS, MySQL, Docker

- A **real-time** collaborative coding platform with multi-user editing, chat, and live code compilation. A web-based IDE(Integrated Development Environment)
- Where I handled overall backend, **docker** to compile code which improved to (~50ms) from **regular API, web socket** for real time messaging in **NodeJs**. Other feature includes **Authentication**, **OTP** code and uses **MySQL** for database.

[Torrex](#) | Rust

- Built a BitTorrent client from scratch in **Rust**, implementing Bencode-based **.torrent parsing**, magnet links via Extended Metadata Exchange, and HTTP tracker-based peer discovery.
- Implemented concurrent peer connections for piece downloading and **SHA-1 hash verification**, with support for pause, resume, and stop operations.
- Developed a REST API using **Actix Web** and **WebSocket** streaming for real-time download progress and client-side download management.

[Aether](#) | Python, Rust, WebRTC, Websocket

- A browser-based remote desktop sharing platform. A end-to-end system for screen capturing and streaming using **webRTC**.
- Backend coordination server manages session signaling, peer connections, and room state video transfer and reflecting mouse pointer.

[dlim-rust](#) | Rust

- A TUI-based asynchronous download manager leveraging IPC (Unix Domain Socket) for inter-process communication.
- Built with **ratatui** for the terminal interface and **tokio** for async runtime. Features concurrent multi-link downloads, pause/resume functionality, and async task monitoring with **tokio-console**.